Proof

$y^2 = 4ax$ is the given parabola. Let (α, β) be the given point.

Equation of the normal in parametric form is

$$y = -tx + 2at + at^3 \quad \text{... (1)}$$

If m is the slope of the normal then $m = -t$.

Therefore the equation (1) becomes $y = mx - 2am - am^3$.

Let it passes through (α, β), then

$$\beta = m\alpha - 2am - am^3$$

$$am^3 + (2a - \alpha)m + \beta = 0$$

which being a cubic equation in m, has three values of m. Consequently three normals, in general, can be drawn from a point to the parabola, since complex roots of real equation, always occur in conjugate pairs and (1) being an odd degree equation, it has atleast one real root. Hence atleast one normal to the parabola is real. ■

5.6.2 Equations of tangent and normal to Ellipse and Hyperbola (the proof of the following are left to the reader)

(1) Equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ cartesian form

(ii) at $'\theta'$ $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$. parametric form

(2) Equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) at (x_1, y_1) is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$ cartesian form

(ii) at $'\theta'$ is $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$ parametric form

(3) Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(i) at (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ cartesian form

(ii) at $'\theta'$ is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$ parametric form

(4) Equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(i) at (x_1, y_1) is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$ cartesian form

(ii) at $'\theta'$ is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$ parametric form.

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